COS 201

WEEK 1–3 PRACTICAL EXERCISES AND PROGRAMMING ASSIGNMENTS

Student Name	Olubodun Oluwapelumi
Matric Number	25/18502
Programme	Computer Science
Course	COS 201

Prepared in a clear, first-year student style.

WEEK 1: INTRODUCTION TO PROBLEM SOLVING

1. Input–Process–Output (IPO)

Problem	Input	Process	Output
Area of a rectangle	Length, width	Area = length × width	Area
Average of four scores	Four scores	Add scores and divide by 4	Average
Positive, negative or zero	A number	Compare the number with 0	Its type
Price of items	Quantity, unit price	Total = quantity × unit price	Total price
Failed login count	Login attempts/status	Count unsuccessful attempts	Number of failed logins

2. Algorithms

a. Find the sum of two numbers

```
1. Start
2. Input A and B
3. Sum = A + B
4. Display Sum
5. Stop
```

b. Calculate the area of a circle

```
1. Start
2. Input radius R
3. Area = 3.142 × R × R
4. Display Area
5. Stop
```

c. Check pass or fail

```
1. Start
2. Input score
3. If score is 50 or above, display Pass
4. Otherwise, display Fail
5. Stop
```

d. Find the largest of three numbers

```
1. Start
2. Input A, B and C
3. Compare the three numbers
4. Display the largest number
5. Stop
```

e. Calculate total item cost

```
1. Start
2. Input quantity and unit price
3. Total = quantity × unit price
4. Display Total
5. Stop
```

3. Pseudocode and flowchart descriptions

a. Addition of two numbers

```
START
INPUT A, B
SUM = A + B
OUTPUT SUM
STOP
```

Textual flowchart: Start → Input A and B → Calculate Sum → Display Sum → Stop.

b. Pass or fail

```
START
INPUT SCORE
IF SCORE >= 50 THEN
```

```
    OUTPUT "PASS"
ELSE
    OUTPUT "FAIL"
ENDIF
STOP
```

Textual flowchart: Start → Input Score → Is Score ≥ 50? → Yes: Display Pass / No: Display Fail → Stop.

WEEK 2: INTRODUCTION TO JAVA PROGRAMMING

1. HelloWorld.java

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

Output:

```
Hello, World!
```

2. Personalised Java program

```
public class MyDetails {
    public static void main(String[] args) {
        System.out.println("Name: Olubodun Oluwapelumi");
        System.out.println("Matric Number: 25/18502");
        System.out.println("Programme: Computer Science");
    }
}
```

Output:

```
Name: Olubodun Oluwapelumi
Matric Number: 25/18502
Programme: Computer Science
```

3. Difference between print() and println()

print() displays text without moving to a new line. **println()** displays text and then moves to the next line.

4. Correct the syntax error

A Java statement that prints a message must end with a semicolon.

```
System.out.println("Welcome to Java");
```

5. README file

```
# COS 201 Practical

This repository contains my COS 201 Week 1–3 practical exercises.

Student: Olubodun Oluwapelumi
Matric Number: 25/18502
Programme: Computer Science

Language: Java
```

6. Submission message

I have completed my COS 201 practical exercises and programming assignments for Weeks 1–3. The source files and README file are included in my GitHub repository.

WEEK 3: BASIC JAVA PROGRAMMING EXERCISES

1. SimpleCalculator.java

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter first number: ");
        double a = input.nextDouble();
        System.out.print("Enter second number: ");
        double b = input.nextDouble();
        System.out.println("Sum = " + (a + b));
        System.out.println("Difference = " + (a - b));
        System.out.println("Product = " + (a * b));
        if (b != 0) {
            System.out.println("Quotient = " + (a / b));
        } else {
            System.out.println("Cannot divide by zero");
        }
        input.close();
    }
}
```

2. AverageCalculator.java

```
import java.util.Scanner;

public class AverageCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter three scores: ");
        double a = input.nextDouble();
        double b = input.nextDouble();
        double c = input.nextDouble();
        double average = (a + b + c) / 3;
        System.out.println("Average = " + average);
        input.close();
    }
}
```

3. ItemCost.java

```
import java.util.Scanner;

public class ItemCost {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter quantity: ");
        int quantity = input.nextInt();
        System.out.print("Enter unit price: ");
        double price = input.nextDouble();
        double total = quantity * price;
        System.out.println("Total cost = " + total);
        input.close();
    }
}
```

4. Example outputs

```
Simple calculator (10 and 5):
Sum = 15.0
Difference = 5.0
Product = 50.0
Quotient = 2.0

Average calculator (60, 70 and 80):
Average = 70.0

Item cost (quantity 3, unit price 500):
Total cost = 1500.0
```

5. README file

```
# COS 201 Week 3

This folder contains basic Java programs for arithmetic calculations, finding an average, and calculating item

Student: Olubodun Oluwapelumi
Matric Number: 25/18502
Programme: Computer Science
```

6. Final submission checklist

• Create a public GitHub repository for the COS 201 practical work.

• Add a README.md file with your student details and project description.

• Save each Java program using the correct class filename.

• Run each program and check that it works.

• Upload the source files and submit the repository link as instructed by your lecturer.

Important: Use this as a guide and compare it with the exact questions and instructions given by your lecturer. If your lecturer supplied different Week 1 or Week 3 tasks, those exact questions should take priority.